

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Panasonic
Firmware Version: N/A
Model(s): PT-VMW50, PT-VMW50U, PT-VMZ50, PT-VMZ50U, PT-VMZ250T, PT-VMZ60, PT-VMW60, PT-VMZ40, PT-VMZ40U, PT-BMX50, PT-BMX50C

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.16.0000-b013	2.18.0

Version History

Module Version	Date	Notes
1_1_1_0	1/10/2023	- Fixed Volume Step command for Ethernet control. - Changed Lamp Usage to Light Usage - Fixed DeviceID
1_0_2_1	10/16/2019	Fixed Volume command
1_0_2_0	9/13/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`
- DeviceID variable must be set accordingly. Default value is '1'. DeviceID ranges from '1' to '64, and Broadcast. If Device ID is set to 'Broadcast', status is unavailable.
Example: `InterfaceName.DeviceID = '1'`
- For Serial control, it is recommended to have a 0.5 second pacing between all commands.

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='PT-VMW50')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='PT-VMW50')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 4352, Model='PT-VMW50')</code>

Control Commands (Serial)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Auto' ⁵ 'Native' 'V-Fit'	Value 'Normal' 'Full'	Value 'Wide' 'H-Fit'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Auto')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage ⁷	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command ClosedCaption ^{5,6}	Value 'Off' 'CC3'	Value 'CC1' 'CC4'	Value 'CC2'
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'Off')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1' 'HDMI 1' 'Memory Viewer'	Value 'Computer 2' ² 'HDMI 2' 'Digital Link' ¹	Value 'Video' 'Network/USB'
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command Keypad	Value '1' – '6'		
# Keypad example InterfaceName.Set('Keypad', '1')			
Command MenuNavigation	Value 'Up' 'Right' 'Return'	Value 'Down' 'Enter'	Value 'Left' 'Menu'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Up')			

Command OnScreenDisplay	Value 'On'	Value 'Off'	
# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'On')			
Command PictureMode	Value 'Dynamic' 'Black Board'	Value 'Natural' ⁴ 'White Board'	Value 'Standard' 'Cinema' ³
# PictureMode example InterfaceName.Set('PictureMode', 'Dynamic')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Volume	Value 0 to 63 in steps of 1		
# Volume example InterfaceName.Set('Volume', 63)			

¹ Only available for the PT-VMW50, PT-VMW50U, PT-VMZ50, PT-VMZ50U, PT-VMZ250T, PT-VMW60 and PT-VMZ60 models.

² Available when Menu → PROJECTOR SETUP → COMPUTER2 SELECT is set to COMPUTER2 IN.

³ Only available for movie-based signals.

⁴ Only available for still image signals.

⁵ Only available for video signal (NTSC) input.

⁶ Not available when the menu screen is displayed.

⁷ Only available for analog RGB signals.

Status Available (Serial)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio ¹	Value 'Auto' ⁵ 'Native' 'V-Fit'	Value 'Normal' 'Full'	Value 'Wide' 'H-Fit'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute ^{5,6}	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command ClosedCaption	Value 'Off' 'CC3'	Value 'CC1' 'CC4'	Value 'CC2'
# ClosedCaption example InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command FilterUsage	Value Hours		
# FilterUsage example InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			

Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'Computer 1' 'HDMI 1' 'Memory Viewer'	Value 'Computer 2' ² 'HDMI 2' 'Digital Link' ¹	Value 'Video' 'Network/USB'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LightUsage	Value Hours		
Qualifier Key 'Light Source'	Qualifier Value '1' – '2'		
# LightUsage example InterfaceName.Update('LightUsage', {'Light Source': '1'}) Value = InterfaceName.ReadStatus('LightUsage', {'Light Source': '1'}) InterfaceName.SubscribeStatus('LightUsage', None, FeedbackHandler)			
Command OnScreenDisplay	Value 'On'	Value 'Off'	
# OnScreenDisplay example InterfaceName.Update('OnScreenDisplay') Value = InterfaceName.ReadStatus('OnScreenDisplay') InterfaceName.SubscribeStatus('OnScreenDisplay', None, FeedbackHandler)			
Command OperationHours	Value Hours		
# OperationHours example InterfaceName.Update('OperationHours') Value = InterfaceName.ReadStatus('OperationHours') InterfaceName.SubscribeStatus('OperationHours', None, FeedbackHandler)			
Command PictureMode	Value 'Dynamic' 'Black Board'	Value 'Natural' ⁴ 'White Board'	Value 'Standard' 'Cinema' ³
# PictureMode example InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command Volume	Value 0 to 63 in steps of 1		
# Volume example InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

¹ Only available for the PT-VMW50, PT-VMW50U, PT-VMZ50, PT-VMZ50U, PT-VMZ250T, PT-VMW60 and PT-VMZ60 models.

² Available when Menu → PROJECTOR SETUP → COMPUTER2 SELECT is set to COMPUTER2 IN.

³ Only available for movie-based signals.

⁴ Only available for still image signals.

⁵ Only available for video signal (NTSC) input.

⁶ Not available when the menu screen is displayed.

Control Commands (Ethernet)

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AVMute	Value 'On'	Value 'Off'	
# AVMute example InterfaceName.Set('AVMute', 'On')			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'Computer 1' 'HDMI 1' 'Memory Viewer'	Value 'Computer 2' ² 'HDMI 2' 'Digital Link' ¹	Value 'Video' 'Network'
# Input example InterfaceName.Set('Input', 'Computer 1')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VolumeStep	Value 'Up'	Value 'Down'	
# VolumeStep example InterfaceName.Set('VolumeStep', 'Up')			

¹ Only available for the PT-VMW50, PT-VMW50U, PT-VMZ50, PT-VMZ50U, PT-VMZ250T, PT-VMW60 and PT-VMZ60 models.

² Available when Menu → PROJECTOR SETUP → COMPUTER2 SELECT is set to COMPUTER2 IN.

Status Available (Ethernet)

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Value	Value	
AudioMute	'On'	'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command	Value	Value	
AVMute	'On'	'Off'	
# AVMute example InterfaceName.Update('AVMute') Value = InterfaceName.ReadStatus('AVMute') InterfaceName.SubscribeStatus('AVMute', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus example Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus	'Normal'	'Multiple Warnings/Errors'	'Fan Warning'
	'Fan Error'	'Lamp Warning'	'Lamp Error'
	'Temperature Warning'	'Temperature Error'	'Filter Warning'
	'Filter Error'	'Other Warning'	'Other Error'
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command	Value		
FilterUsage	Hours		
# FilterUsage example InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command	Value	Value	
Freeze	'On'	'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			

Command Input	Value 'Computer 1' 'HDMI 1' 'Memory Viewer'	Value 'Computer 2' ² 'HDMI 2' 'Digital Link' ¹	Value 'Video' 'Network'
<pre># Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)</pre>			
Command LightUsage	Value Hours		
<pre># LightUsage example InterfaceName.Update('LightUsage') Value = InterfaceName.ReadStatus('LightUsage') InterfaceName.SubscribeStatus('LightUsage', None, FeedbackHandler)</pre>			
Command Power	Value 'On'	Value 'Off'	Value 'Cooling Down'
<pre># Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			

¹ Only available for the PT-VMW50, PT-VMW50U, PT-VMZ50, PT-VMZ50U, PT-VMZ250T, PT-VMW60 and PT-VMZ60 models.

² Available when Menu → PROJECTOR SETUP → COMPUTER2 SELECT is set to COMPUTER2 IN.

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

To ensure that the module functions correctly, please set the following:

- Menu → PROJECTOR SETUP → ECO MANAGEMENT → STANDBY MODE: NORMAL.
- Menu → PROJECTOR SETUP → RS-232C → PARITY: NONE.
- Menu → PROJECTOR SETUP → RS-232C → EMULATE: OFF.
- Menu → PROJECTOR SETUP → RS-232C → INPUT SELECT: PROJECTOR or DIGITAL LINK, depending on whether communication occurs through the projector or the optional Digital Link output supported device.
 - When communicating through Digital Link, ensure that Menu → NETWORK → DIGITAL LINK MODE is not set to ETHERNET.

Supported baud rates are 9600, 19200, and 38400 bps.

For the PT-VMW50, PT-VMW50U, PT-VMZ50, PT-VMZ50U, PT-VMZ250T, PT-VMW60 and PT-VMZ60 models, the baud rate should be set to 9600 bps when communicating through Digital Link.

Serial communication

Port Type: RS-232

Baud Rate: 9600

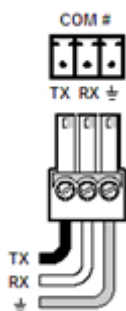
Data Bits: 8

Parity: None

Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	↗	2	TxD
RxD	↖	3	RxD
GND	—	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet (TCP)

Default Port: 4352

Logon Credentials Supported: Yes

Default Password: @Panasonic

Multi-Connection Capabilities: Undetermined

Port Changeability: No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

To ensure the functionality of the module, please set the following menu options:

- Menu → PROJECTOR SETUP → ECO MANAGEMENT → STANDBY MODE: NORMAL
- Menu → NETWORK → NETWORK CONTROL → PJLink CONTROL: ON

The password used for ethernet control is the same password set for web control. If you wish to use ethernet control without a password, disable the password for web control.

Appendix A. Set Commands(Serial, Device ID: 1)

Aspect Ratio Auto	\x02AD01;VSE:0\x03
Aspect Ratio Full	\x02AD01;VSE:6\x03
Aspect Ratio H-Fit	\x02AD01;VSE:9\x03
Aspect Ratio Native	\x02AD01;VSE:5\x03
Aspect Ratio Normal	\x02AD01;VSE:1\x03
Aspect Ratio V-Fit	\x02AD01;VSE:10\x03
Aspect Ratio Wide	\x02AD01;VSE:2\x03
Audio Mute Off	\x02AD01;AMT:0\x03
Audio Mute On	\x02AD01;AMT:1\x03
Auto Image None	\x02AD01;OAS\x03
AV Mute Off	\x02AD01;OSH:0\x03
AV Mute On	\x02AD01;OSH:1\x03
Closed Caption CC1	\x02AD01;OCC:1\x03
Closed Caption CC2	\x02AD01;OCC:2\x03
Closed Caption CC3	\x02AD01;OCC:3\x03
Closed Caption CC4	\x02AD01;OCC:4\x03
Closed Caption Off	\x02AD01;OCC:0\x03
Freeze Off	\x02AD01;OFZ:0\x03
Freeze On	\x02AD01;OFZ:1\x03
Input Computer 1	\x02AD01;IIS:RG1\x03
Input Computer 2	\x02AD01;IIS:RG2\x03
Input Digital Link	\x02AD01;IIS:DL1\x03
Input HDMI 1	\x02AD01;IIS:HD1\x03
Input HDMI 2	\x02AD01;IIS:HD2\x03
Input Memory Viewer	\x02AD01;IIS:MV1\x03
Input Network/USB	\x02AD01;IIS:NWP\x03
Input Video	\x02AD01;IIS:VID\x03
Keypad 1	\x02AD01;ONK:1\x03
Keypad 6	\x02AD01;ONK:6\x03
Menu Navigation Down	\x02AD01;OCD\x03
Menu Navigation Enter	\x02AD01;OEN\x03
Menu Navigation Left	\x02AD01;OCL\x03
Menu Navigation Menu	\x02AD01;OMN\x03
Menu Navigation Return	\x02AD01;OBK\x03
Menu Navigation Right	\x02AD01;OCR\x03
Menu Navigation Up	\x02AD01;OCU\x03
On Screen Display Off	\x02AD01;OOS:0\x03
On Screen Display On	\x02AD01;OOS:1\x03
Picture Mode Black Board	\x02AD01;VPM:BBD\x03
Picture Mode Cinema	\x02AD01;VPM:CIN\x03
Picture Mode Dynamic	\x02AD01;VPM:DYN\x03
Picture Mode Natural	\x02AD01;VPM:NAT\x03

Global Scriptor Module Communication Sheet

Revision: 1/10/2023

Picture Mode Standard	\x02AD01;VPM:STD\x03
Picture Mode White Board	\x02AD01;VPM:WBD\x03
Power Off	\x02AD01;POF\x03
Power On	\x02AD01;PON\x03
Volume 0	\x02AD01;AVL:000\x03
Volume 63	\x02AD01;AVL:063\x03

Appendix B. Update Commands(Serial, Device ID: 1)

Aspect Ratio	\x02AD01;QSE\x03
Audio Mute	\x02AD01;QMT\x03
AV Mute	\x02AD01;QSH\x03
Closed Caption	\x02AD01;QCC\x03
Filter Usage	\x02AD01;QFI:0\x03
Freeze	\x02AD01;QFZ\x03
Input	\x02AD01;QIN\x03
Light Usage Light Source 1	\x02AD01;QVX:LRTS3=00\x03
Light Usage Light Source 2	\x02AD01;QVX:LRTS3=01\x03
On Screen Display	\x02AD01;QOS\x03
Operation Hours	\x02AD01;QVX:RTMI0\x03
Picture Mode	\x02AD01;QPM\x03
Power	\x02AD01;QPW\x03
Volume	\x02AD01;QAV\x03

Appendix C. Set Commands(Ethernet)

Audio Mute Off	%1AVMT 20\x0D
Audio Mute On	%1AVMT 21\x0D
AV Mute Off	%1AVMT 30\x0D
AV Mute On	%1AVMT 31\x0D
Freeze Off	%2FREZ 0\x0D
Freeze On	%2FREZ 1\x0D
Input Computer 1	%1INPT 11\x0D
Input Computer 2	%1INPT 12\x0D
Input Digital Link	%1INPT 33\x0D
Input HDMI 1	%1INPT 31\x0D
Input HDMI 2	%1INPT 32\x0D
Input Memory Viewer	%1INPT 41\x0D
Input Network	%1INPT 51\x0D
Input Video	%1INPT 21\x0D
Power Off	%1POWR 0\x0D
Power On	%1POWR 1\x0D
Volume Step Down	%2SVOL 0\x0D
Volume Step Up	%2SVOL 1\x0D

Appendix D. Update Commands(Ethernet)

Audio Mute	%1AVMT ?\x0D
Device Status	%1ERST ?\x0D
Filter Usage	%2FILT ?\x0D
Freeze	%2FREZ ?\x0D
Input	%1INPT ?\x0D
Light Usage	%1LAMP ?\x0D
Light Usage Light Source 1	%1LAMP ?\x0D
Light Usage Light Source 2	%1LAMP ?\x0D
Power	%1POWR ?\x0D